

Early Warning Risiko Mahasiswa OULAD

Evaluasi Model, Knowledge-Based System, dan Dashboard DVBI

Mata Kuliah: Data Visualization and Business Intelligence

Kelompok: 5

Program Studi: S2 PJJ Informatika, Konsentrasi Big Data dan Predictive Analytics

Institusi: Universitas Amikom Yogyakarta

Nama	NIM
Muhammad Rizky Hajar	24.55.2714
Alwie Muflich	24.55.2667
Heri Santosa	24.55.2676

Tujuan notebook: memprediksi mahasiswa berisiko pada akhir minggu ke-4, membandingkan tiga model supervised learning, memilih model dengan *recall* `AtRisk` terbaik, lalu menerjemahkan hasilnya menjadi sistem berbasis pengetahuan dan dashboard pendukung keputusan.

Definisi target: `AtRisk` = `Withdrawn` atau `Fail`; `Successful` = `Pass` atau `Distinction`.

Hasil model berfungsi sebagai **decision support**. Dosen, tutor, dan pengelola akademik menetapkan keputusan intervensi berdasarkan hasil analisis dan pertimbangan akademik.

1. Persiapan library

Tujuan: menyiapkan seluruh library dan parameter eksperimen.

Langkah: mengimpor library pengolahan data, visualisasi, preprocessing, evaluasi, dan tiga algoritma klasifikasi. `RANDOM_STATE=42` menjaga hasil tetap dapat direproduksi, sedangkan `CUTOFF_DAY=28` membatasi data hingga akhir minggu ke-4.

Output: pesan konfigurasi eksperimen.

```
import os, shutil, urllib.request, zipfile, warnings
from pathlib import Path
from collections import Counter

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.base import clone
from sklearn.compose import ColumnTransformer
from sklearn.ensemble import RandomForestClassifier
from sklearn.impute import SimpleImputer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (accuracy_score, classification_report, confusion_matrix,
                             f1_score, precision_score, recall_score, roc_auc_score,
                             roc_curve)
from sklearn.model_selection import GroupKFold, GroupShuffleSplit
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from xgboost import XGBClassifier

warnings.filterwarnings('ignore')
sns.set_theme(style='whitegrid', palette='Set2')
pd.set_option('display.max_columns', 100)
```

```

RANDOM_STATE = 42
CUTOFF_DAY = 28
DATA_DIR = Path('/content/oulad')
print(f'Konfigurasi: random_state={RANDOM_STATE}, cutoff={CUTOFF_DAY} hari')

```

Konfigurasi: random_state=42, cutoff=28 hari

2. Mengunduh dataset OULAD

Tujuan: memperoleh data tanpa upload manual atau Google Drive.

Langkah: mengunduh arsip OULAD dari UCI Machine Learning Repository, mengekstraknya, dan memastikan tujuh file sumber tersedia. Download dilewati bila file sudah ada dalam runtime.

Output: daftar file CSV dan status validasi.

```

DATA_URL = 'https://archive.ics.uci.edu/static/public/349/open+university+learning+analytics+dataset.zip'
ZIP_PATH = Path('/content/anonymisedData.zip')
EXPECTED_FILES = {'assessments.csv', 'courses.csv', 'studentAssessment.csv',
                  'studentInfo.csv', 'studentRegistration.csv',
                  'studentVle.csv', 'vle.csv'}

DATA_DIR.mkdir(parents=True, exist_ok=True)
available = {p.name for p in DATA_DIR.glob('*.csv')}
if not EXPECTED_FILES.issubset(available):
    print('Mengunduh OULAD...')
    urllib.request.urlretrieve(DATA_URL, ZIP_PATH)
    with zipfile.ZipFile(ZIP_PATH) as archive:
        archive.extractall(DATA_DIR)

available = {p.name for p in DATA_DIR.glob('*.csv')}
missing = EXPECTED_FILES - available
assert not missing, f'File dataset tidak lengkap: {sorted(missing)}'
print('Dataset siap:', ', '.join(sorted(EXPECTED_FILES)))

```

Dataset siap: assessments.csv, courses.csv, studentAssessment.csv, studentInfo.csv, studentRegistration.csv, studentVle.csv, vle.csv

3. Membaca dan memeriksa data sumber

Tujuan: memahami ukuran dan struktur tabel yang dipakai.

Langkah: membaca empat tabel utama. `studentInfo` memuat profil dan hasil akhir, `studentRegistration` memuat tanggal registrasi, `assessments` memberi konteks modul, dan dua tabel aktivitas menyediakan perilaku assessment serta VLE.

Output: jumlah baris dan kolom setiap tabel.

```

student_info = pd.read_csv(DATA_DIR / 'studentInfo.csv')
registration = pd.read_csv(DATA_DIR / 'studentRegistration.csv')
assessments = pd.read_csv(DATA_DIR / 'assessments.csv')
student_assessment = pd.read_csv(DATA_DIR / 'studentAssessment.csv')
student_vle = pd.read_csv(DATA_DIR / 'studentVle.csv')

source_summary = pd.DataFrame({
    'tabel': ['studentInfo', 'studentRegistration', 'assessments', 'studentAssessment', 'studentVle'],
    'baris': [len(student_info), len(registration), len(assessments), len(student_assessment), len(student_vle)],
    'kolom': [student_info.shape[1], registration.shape[1], assessments.shape[1],
              student_assessment.shape[1], student_vle.shape[1]]
})
display(source_summary.style.format(thousands=','))

```

	tabel	baris	kolom
0	studentInfo	32,593	12
1	studentRegistration	32,593	5
2	assessments	206	6
3	studentAssessment	173,912	5
4	studentVle	10,655,280	6

4. Membentuk dataset early warning minggu ke-4

Tujuan: menghasilkan fitur yang hanya tersedia sampai hari ke-28.

Langkah: assessment difilter memakai `date_submitted <= 28`, sedangkan aktivitas VLE difilter memakai `date <= 28`. Data kemudian diagregasi per mahasiswa, modul, dan presentation. Fitur model berfokus pada informasi yang tersedia hingga hari ke-28; tanggal unregistration dikeluarkan karena bersifat post-hoc.

Output: ukuran dataset, jumlah mahasiswa unik, dan daftar fitur.

```
KEYS = ['code_module', 'code_presentation', 'id_student']

# Target biner berasal dari hasil akhir; fitur prediktor menggunakan data yang tersedia sampai hari ke-28.
base = student_info.copy()
base['risk_label'] = np.where(base['final_result'].isin(['Withdrawn', 'Fail']), 'AtRisk', 'Successful')

# Fitur registrasi yang tersedia sebelum/awal perkuliahan; unregistration sengaja dikeluarkan.
reg_features = registration[KEYS + ['date_registration']].copy()

# Hubungkan submission ke konteks module-presentation, lalu batasi sampai hari ke-28.
student_assessment['date_submitted'] = pd.to_numeric(student_assessment['date_submitted'], errors='coerce')
student_assessment['score'] = pd.to_numeric(student_assessment['score'], errors='coerce')
assessment_early = student_assessment.merge(
    assessments[['id_assessment', 'code_module', 'code_presentation']],
    on='id_assessment', how='left'
)
assessment_early = assessment_early[assessment_early['date_submitted'] <= CUTOFF_DAY]
assessment_features = (assessment_early.groupby(KEYS)
    .agg(assessment_count=('id_assessment', 'count'),
        assessment_score_mean=('score', 'mean'),
        assessment_score_max=('score', 'max'),
        assessment_score_min=('score', 'min'))
    .reset_index())

# Aktivitas VLE juga dibatasi sampai hari ke-28.
student_vle['date'] = pd.to_numeric(student_vle['date'], errors='coerce')
student_vle['sum_click'] = pd.to_numeric(student_vle['sum_click'], errors='coerce').fillna(0)
vle_early = student_vle[student_vle['date'] <= CUTOFF_DAY].copy()
vle_features = (vle_early.groupby(KEYS)
    .agg(vle_total_clicks=('sum_click', 'sum'),
        vle_active_days=('date', 'nunique'),
        vle_site_count=('id_site', 'nunique'),
        vle_last_activity_day=('date', 'max'))
    .reset_index())

dataset = (base.merge(reg_features, on=KEYS, how='left')
    .merge(assessment_features, on=KEYS, how='left')
    .merge(vle_features, on=KEYS, how='left'))

behavior_cols = ['assessment_count', 'assessment_score_mean', 'assessment_score_max',
    'assessment_score_min', 'vle_total_clicks', 'vle_active_days',
    'vle_site_count', 'vle_last_activity_day']
dataset[behavior_cols] = dataset[behavior_cols].fillna(0)

print(f'Unit analisis: {len(dataset):,} student-module-presentation')
print(f'Mahasiswa unik: {dataset.id_student.nunique():,}')
display(dataset.head())
```

Unit analisis: 32,593 student-module-presentation
Mahasiswa unik: 28,785

	code module	code presentation	id_ student	gender	region	highest education	imd_ band	age_ band	num_ of_ prev_ attempts	studied_ credits	disability	final_ result	risk_ label	date_ registration	assessment_ count	assessment_ score_ mean	assessment_ score_ max	assessment_ score_ min	vle_ total_ clicks	vle_ active_ days	vle_ site_ count	vle_ last_ activity_ day
0	AAA	2013 J	1139 1	M	East Anglian Region	HE Qualification	90-1 00%	55<=	0	240	N	Pass	Successful	-159. 0	1.0	78.0	78.0	78.0	401.0	8.0	24.0	18.0
1	AAA	2013 J	2840 0	F	Scotland	HE Qualification	20-3 0%	35-5 5	0	60	N	Pass	Successful	-53.0	1.0	70.0	70.0	70.0	618.0	19.0	34.0	28.0
2	AAA	2013 J	3026 8	F	North West ern Region	A Level or Equivalent	30-4 0%	35-5 5	0	60	Y	Withdrawn	AtRisk	-92.0	0.0	0.0	0.0	0.0	281.0	12.0	22.0	12.0
3	AAA	2013 J	3160 4	F	South East Region	A Level or Equivalent	50-6 0%	35-5 5	0	60	N	Pass	Successful	-52.0	1.0	72.0	72.0	72.0	494.0	22.0	31.0	28.0
4	AAA	2013 J	3288 5	F	West Midlands Region	Lower Than A Level	50-6 0%	0-35	0	60	N	Pass	Successful	-176. 0	1.0	69.0	69.0	69.0	567.0	24.0	34.0	26.0

5. Validasi anti-leakage

Tujuan: memastikan seluruh fitur prediktor tersedia pada atau sebelum hari ke-28.

Langkah: mendefinisikan fitur model secara eksplisit dan memastikan daftar prediktor terbatas pada data demografis, registrasi awal, assessment awal, dan aktivitas VLE awal.

Output: status `LULUS` dan daftar fitur kategorikal/numerik.

```
CATEGORICAL_COLS = ['gender', 'region', 'highest_education', 'imd_band',
                    'age_band', 'disability', 'code_module', 'code_presentation']
NUMERIC_COLS = ['num_of_prev_attempts', 'studied_credits', 'date_registration',
                'assessment_count', 'assessment_score_mean', 'assessment_score_max',
                'assessment_score_min', 'vle_total_clicks', 'vle_active_days',
                'vle_site_count', 'vle_last_activity_day']
FEATURE_COLS = CATEGORICAL_COLS + NUMERIC_COLS
FORBIDDEN = {'final_result', 'risk_label', 'date_unregistration', 'has_unregistration'}

assert not FORBIDDEN.intersection(FEATURE_COLS)
assert vle_early['date'].max() <= CUTOFF_DAY
assert assessment_early['date_submitted'].max() <= CUTOFF_DAY
print('LULUS: tidak ada fitur target, unregistration, atau aktivitas setelah hari ke-28.')
print('Fitur kategorikal:', CATEGORICAL_COLS)
print('Fitur numerik:', NUMERIC_COLS)
```

LULUS: tidak ada fitur target, unregistration, atau aktivitas setelah hari ke-28.
 Fitur kategorikal: ['gender', 'region', 'highest_education', 'imd_band', 'age_band', 'disability', 'code_module', 'code_presentation']
 Fitur numerik: ['num_of_prev_attempts', 'studied_credits', 'date_registration', 'assessment_count', 'assessment_score_mean', 'assessment_score_max', 'assessment_score_min', 'vle_total_clicks', 'vle_active_days', 'vle_site_count', 'vle_last_activity_day']

6. Exploratory Data Analysis

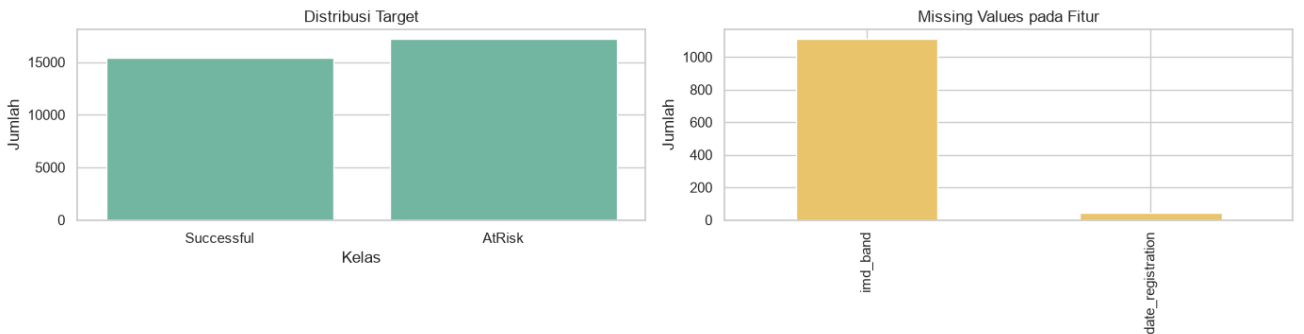
Tujuan: melihat kualitas data dan pola awal risiko.

Langkah: menampilkan distribusi kelas, missing values, dan median perilaku minggu ke-4 untuk kedua kelas.

Output: keseimbangan target serta perbedaan assessment dan VLE antara `AtRisk` dan `Successful`.

```
missing_summary = dataset[FEATURE_COLS].isna().sum().sort_values(ascending=False)
class_counts = dataset['risk_label'].value_counts()
behavior_by_class = dataset.groupby('risk_label')[
    ['assessment_count', 'assessment_score_mean', 'vle_total_clicks', 'vle_active_days']]
    .median().round(2)
```

```
fig, axes = plt.subplots(1, 2, figsize=(14, 4))
sns.countplot(data=dataset, x='risk_label', order=['Successful', 'AtRisk'], ax=axes[0])
axes[0].set_title('Distribusi Target')
axes[0].set_xlabel('Kelas'); axes[0].set_ylabel('Jumlah')
missing_nonzero = missing_summary[missing_summary > 0]
if missing_nonzero.empty:
    axes[1].text(0.5, 0.55, 'Kelengkapan fitur: 100%', ha='center', va='center',
                fontsize=18, fontweight='bold', color='#2A9D8F')
    axes[1].text(0.5, 0.40, 'Seluruh fitur siap digunakan', ha='center', va='center')
    axes[1].set_axis_off()
else:
    missing_nonzero.plot(kind='bar', ax=axes[1], color='#E9C46A')
    axes[1].set_title('Missing Values pada Fitur')
    axes[1].set_xlabel('Fitur'); axes[1].set_ylabel('Jumlah')
plt.tight_layout(); plt.show()
display(behavior_by_class)
print(class_counts)
```



	assessment_count	assessment_score_mean	vle_total_clicks	vle_active_days
AtRisk	1.0	0.0	92.0	7.0
Successful	1.0	76.0	298.0	16.0

```
risk_label
AtRisk      17208
Successful  15385
Name: count, dtype: int64
```

7. Pembagian data berdasarkan mahasiswa

Tujuan: membuat evaluasi hold-out yang adil.

Langkah: membagi 80% train-validation dan 20% test menggunakan `id_student` sebagai grup. Setiap mahasiswa ditempatkan secara eksklusif pada salah satu bagian walaupun mengambil lebih dari satu modul.

Output: ukuran split dan jumlah overlap mahasiswa yang harus bernilai nol.

```
for column in NUMERIC_COLS:
    dataset[column] = pd.to_numeric(dataset[column], errors='coerce')

X = dataset[FEATURE_COLS].copy()
y = dataset['risk_label'].map({'Successful': 0, 'AtRisk': 1})
groups = dataset['id_student']

splitter = GroupShuffleSplit(n_splits=1, test_size=0.20, random_state=RANDOM_STATE)
train_idx, test_idx = next(splitter.split(X, y, groups))
X_train, X_test = X.iloc[train_idx].copy(), X.iloc[test_idx].copy()
y_train, y_test = y.iloc[train_idx].copy(), y.iloc[test_idx].copy()
groups_train = groups.iloc[train_idx].copy()

train_students = set(groups.iloc[train_idx])
test_students = set(groups.iloc[test_idx])
overlap = train_students.intersection(test_students)
assert len(overlap) == 0
print(f'Train-validation: {len(X_train):,} baris | Test: {len(X_test):,} baris')
print(f'Overlap mahasiswa: {len(overlap):,}')
print('Distribusi train:', y_train.value_counts(normalize=True).round(3).to_dict())
print('Distribusi test :', y_test.value_counts(normalize=True).round(3).to_dict())
```

```
Train-validation: 26,122 baris | Test: 6,471 baris
Overlap mahasiswa: 0
Distribusi train: {1: 0.529, 0: 0.471}
Distribusi test : {1: 0.525, 0: 0.475}
```

8. Pipeline preprocessing dan model

Tujuan: mendefinisikan transformasi dan tiga model secara konsisten.

Langkah: nilai numerik diimputasi dengan median dan distandardisasi; kategori diimputasi dan diubah dengan one-hot encoding. Semua transformasi berada di dalam pipeline sehingga hanya belajar dari data train pada setiap fold.

Output: nama tiga model pembandingan.

```
numeric_pipe = Pipeline([
    ('imputer', SimpleImputer(strategy='median')),
    ('scaler', StandardScaler())
])
categorical_pipe = Pipeline([
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('onehot', OneHotEncoder(handle_unknown='ignore'))
])
preprocessor = ColumnTransformer([
    ('numeric', numeric_pipe, NUMERIC_COLS),
    ('categorical', categorical_pipe, CATEGORICAL_COLS)
])

negative, positive = np.bincount(y_train)
scale_pos_weight = negative / positive
models = {
    'Logistic Regression': LogisticRegression(max_iter=1500, class_weight='balanced', random_state=RANDOM_STATE),
    'Random Forest': RandomForestClassifier(n_estimators=250, class_weight='balanced',
                                           random_state=RANDOM_STATE, n_jobs=-1),
    'XGBoost': XGBClassifier(n_estimators=200, max_depth=4, learning_rate=0.08,
                             subsample=0.9, colsample_bytree=0.9,
                             scale_pos_weight=scale_pos_weight, eval_metric='logloss',
                             random_state=RANDOM_STATE, n_jobs=-1)
}
pipelines = {name: Pipeline([('preprocess', clone(preprocessor)), ('model', model)])
              for name, model in models.items()}
print('Model:', ', '.join(pipelines))
```

Model: Logistic Regression, Random Forest, XGBoost

9. Cross-validation lima fold

Tujuan: mengukur kestabilan model pada beberapa pembagian data latih.

Langkah: setiap model dievaluasi dengan lima fold berbasis kelompok mahasiswa. Metrik utama adalah recall `AtRisk`; F1 digunakan sebagai tie-breaker.

Output: rata-rata dan standar deviasi accuracy, precision, recall, F1, dan ROC-AUC.

```
def calculate_metrics(y_true, y_pred, y_prob):
    return {
        'accuracy': accuracy_score(y_true, y_pred),
        'precision_atrisk': precision_score(y_true, y_pred, zero_division=0),
        'recall_atrisk': recall_score(y_true, y_pred, zero_division=0),
        'f1_atrisk': f1_score(y_true, y_pred, zero_division=0),
        'roc_auc': roc_auc_score(y_true, y_prob)
    }

gkf = GroupKFold(n_splits=5)
cv_rows = []
for model_name, pipeline in pipelines.items():
    for fold, (fit_idx, val_idx) in enumerate(gkf.split(X_train, y_train, groups_train), start=1):
        fold_model = clone(pipeline)
        fold_model.fit(X_train.iloc[fit_idx], y_train.iloc[fit_idx])
        pred = fold_model.predict(X_train.iloc[val_idx])
        prob = fold_model.predict_proba(X_train.iloc[val_idx])[:, 1]
        cv_rows.append({'model': model_name, 'fold': fold,
                        **calculate_metrics(y_train.iloc[val_idx], pred, prob)})

cv_detail = pd.DataFrame(cv_rows)
cv_summary = cv_detail.groupby('model').agg(
    accuracy_mean=('accuracy', 'mean'), accuracy_std=('accuracy', 'std'),
    precision_mean=('precision_atrisk', 'mean'), precision_std=('precision_atrisk', 'std'),
```

```

recall_mean=('recall_atrisk', 'mean'), recall_std=('recall_atrisk', 'std'),
f1_mean=('f1_atrisk', 'mean'), f1_std=('f1_atrisk', 'std'),
roc_auc_mean=('roc_auc', 'mean'), roc_auc_std=('roc_auc', 'std')
).sort_values(['recall_mean', 'f1_mean'], ascending=False)
display(cv_summary.style.format('{:.4f}').highlight_max(subset=['recall_mean', 'f1_mean'], color='#B7E4C7'))

```

model	accuracy_mean	accuracy_std	precision_mean	precision_std	recall_mean	recall_std	f1_mean	f1_std	roc_auc_mean	roc_auc_std
Random Forest	0.7538	0.0033	0.7999	0.0148	0.7126	0.0040	0.7536	0.0050	0.8362	0.0026
XGBoost	0.7582	0.0041	0.8217	0.0093	0.6931	0.0033	0.7519	0.0028	0.8440	0.0020
Logistic Regression	0.7484	0.0033	0.8081	0.0063	0.6871	0.0038	0.7427	0.0043	0.8324	0.0028

10. Evaluasi akhir pada hold-out test

Tujuan: membandingkan generalisasi ketiga model pada data yang belum pernah dilihat.

Langkah: model dipilih dari hasil cross-validation berdasarkan recall dan F1 `AtRisk`, kemudian dilatih pada seluruh train-validation. Hold-out test digunakan untuk mengukur generalisasi akhir dan menampilkan perbandingan model.

Output: tabel metrik dan nama model terbaik.

```

fitted_models, test_predictions, test_probabilities, test_rows = {}, {}, {}, []
for model_name, pipeline in pipelines.items():
    fitted = clone(pipeline).fit(X_train, y_train)
    pred = fitted.predict(X_test)
    prob = fitted.predict_proba(X_test)[:, 1]
    fitted_models[model_name] = fitted
    test_predictions[model_name] = pred
    test_probabilities[model_name] = prob
    test_rows.append({'model': model_name, **calculate_metrics(y_test, pred, prob)})

test_results = pd.DataFrame(test_rows).set_index('model').sort_values(
    ['recall_atrisk', 'f1_atrisk'], ascending=False)
best_model_name = cv_summary.index[0]
best_model = fitted_models[best_model_name]
best_pred = test_predictions[best_model_name]
best_prob = test_probabilities[best_model_name]

display(test_results.style.format('{:.4f}').highlight_max(subset=['recall_atrisk', 'f1_atrisk'], color='#B7E4C7'))
print(f'Model terbaik: {best_model_name}')
print(classification_report(y_test, best_pred, target_names=['Successful', 'AtRisk']))

```

model	accuracy	precision_atrisk	recall_atrisk	f1_atrisk	roc_auc
Random Forest	0.7592	0.8032	0.7172	0.7578	0.8396
XGBoost	0.7633	0.8186	0.7054	0.7578	0.8440
Logistic Regression	0.7476	0.8040	0.6869	0.7408	0.8298

```

Model terbaik: Random Forest
precision    recall  f1-score   support

Successful   0.72    0.81    0.76     3073
AtRisk       0.80    0.72    0.76     3398

accuracy
macro avg   0.76    0.76    0.76     6471
weighted avg 0.76    0.76    0.76     6471

```

11. Visualisasi evaluasi model

Tujuan: membuat performa model mudah dibandingkan secara visual.

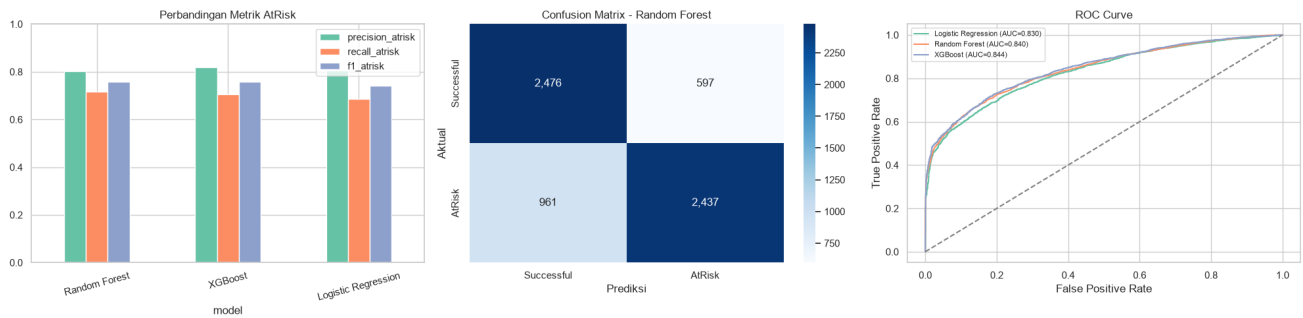
Langkah: menampilkan perbandingan metrik, confusion matrix model terbaik, dan ROC curve.

Output: false negative pada confusion matrix karena bagian ini menunjukkan mahasiswa `AtRisk` yang terlewat.

```
fig, axes = plt.subplots(1, 3, figsize=(20, 5))
test_results[['precision_atrisk', 'recall_atrisk', 'f1_atrisk']].plot(kind='bar', ax=axes[0])
axes[0].set_title('Perbandingan Metrik AtRisk'); axes[0].set_ylim(0, 1); axes[0].tick_params(axis='x',
rotation=15)

cm_best = confusion_matrix(y_test, best_pred)
sns.heatmap(cm_best, annot=True, fmt='d', cmap='Blues', ax=axes[1],
            xticklabels=['Successful', 'AtRisk'], yticklabels=['Successful', 'AtRisk'])
axes[1].set_title(f'Confusion Matrix - {best_model_name}')
axes[1].set_xlabel('Prediksi'); axes[1].set_ylabel('Aktual')

for name, prob in test_probabilities.items():
    fpr, tpr, _ = roc_curve(y_test, prob)
    axes[2].plot(fpr, tpr, label=f"{name} (AUC={roc_auc_score(y_test, prob):.3f})")
axes[2].plot([0, 1], [0, 1], '--', color='gray')
axes[2].set_title('ROC Curve'); axes[2].set_xlabel('False Positive Rate'); axes[2].set_ylabel('True Positive Rate')
axes[2].legend(fontsize=8)
plt.tight_layout(); plt.show()
```



12. Faktor yang memengaruhi model terbaik

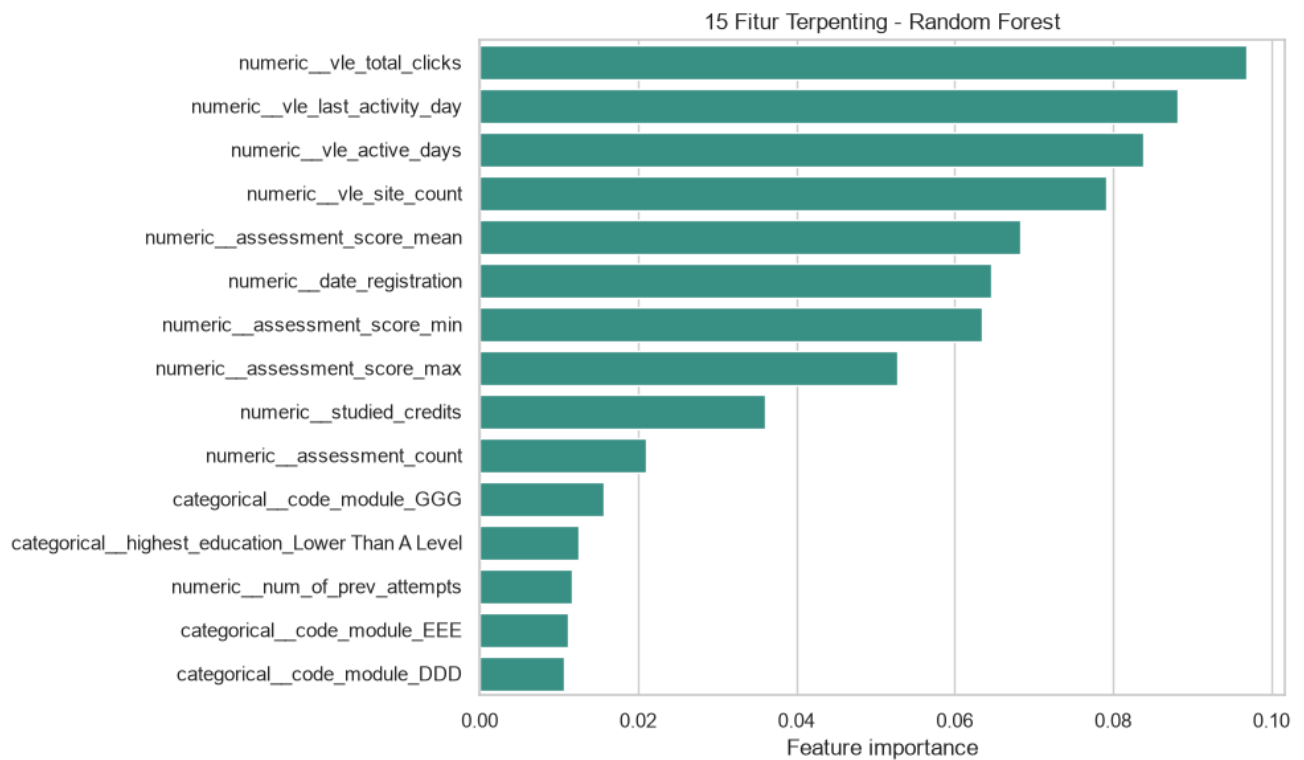
Tujuan: memberi transparansi mengenai fitur yang paling berpengaruh.

Langkah: mengambil nama fitur setelah preprocessing, kemudian menampilkan feature importance untuk model pohon atau nilai absolut coefficient untuk Logistic Regression.

Output: 15 fitur dengan pengaruh terbesar.

```
feature_names = best_model.named_steps['preprocess'].get_feature_names_out()
estimator = best_model.named_steps['model']
if hasattr(estimator, 'feature_importances_'):
    importance_values = estimator.feature_importances_
    importance_label = 'Feature importance'
else:
    importance_values = np.abs(estimator.coef_[0])
    importance_label = 'Absolute coefficient'

feature_importance = (pd.DataFrame({'feature': feature_names, 'importance': importance_values})
                      .sort_values('importance', ascending=False).head(15))
plt.figure(figsize=(10, 6))
sns.barplot(data=feature_importance, x='importance', y='feature', color='#2A9D8F')
plt.title(f'15 Fitur Terpenting - {best_model_name}')
plt.xlabel(importance_label); plt.ylabel(''); plt.tight_layout(); plt.show()
display(feature_importance)
```



	feature	importance
7	numeric__vle_total_clicks	0.096883
10	numeric__vle_last_activity_day	0.088235
8	numeric__vle_active_days	0.083810
9	numeric__vle_site_count	0.079148
4	numeric__assessment_score_mean	0.068352
2	numeric__date_registration	0.064697
6	numeric__assessment_score_min	0.063415
5	numeric__assessment_score_max	0.052808
1	numeric__studied_credits	0.036006
3	numeric__assessment_count	0.020966
52	categorical__code_module_GGG	0.015661
28	categorical__highest_education_Lower Than A Level	0.012509
0	numeric__num_of_prev_attempts	0.011603
50	categorical__code_module_EEE	0.011123
49	categorical__code_module_DDD	0.010725

13. Knowledge-based risk layer

Tujuan: mengubah prediksi model menjadi kategori dan alasan yang operasional.

Langkah: threshold perilaku dihitung hanya dari data train menggunakan kuartil bawah. Mahasiswa diberi sinyal risiko, kemudian probabilitas model dan jumlah sinyal digabungkan menjadi **High**, **Medium**, atau **Low Risk**.

Aturan: High = model **AtRisk** dan minimal dua sinyal; Medium = model **AtRisk** atau minimal dua sinyal; selain itu Low.

Output: threshold, distribusi level, alasan, dan rekomendasi.

```

RULE_COLS = ['assessment_score_mean', 'assessment_count', 'vle_total_clicks', 'vle_active_days']
thresholds = X_train[RULE_COLS].quantile(0.25).to_dict()

def add_knowledge_layer(frame, ml_pred, ml_prob, thresholds):
    result = frame.copy()
    result['predicted_atrisk'] = np.asarray(ml_pred).astype(int)
    result['probability_atrisk'] = np.asarray(ml_prob)
    signal_map = {
        'Skor assessment rendah': result['assessment_score_mean'] <= thresholds['assessment_score_mean'],
        'Partisipasi assessment rendah': result['assessment_count'] <= thresholds['assessment_count'],
        'Total klik VLE rendah': result['vle_total_clicks'] <= thresholds['vle_total_clicks'],
        'Hari aktif VLE rendah': result['vle_active_days'] <= thresholds['vle_active_days']
    }

```

```

}
result['risk_signal_count'] = sum(condition.astype(int) for condition in signal_map.values())
result['risk_reasons'] = [
    '; '.join(name for name, condition in signal_map.items() if bool(condition.iloc[i])) or 'Tidak ada sinyal
aturan'
    for i in range(len(result))
]
high = (result['predicted_atrisk'] == 1) & (result['risk_signal_count'] >= 2)
medium = (result['predicted_atrisk'] == 1) | (result['risk_signal_count'] >= 2)
result['knowledge_risk_level'] = np.select([high, medium], ['High Risk', 'Medium Risk'], default='Low Risk')

def recommendation(row):
    reasons = row['risk_reasons']
    actions = []
    if 'VLE' in reasons: actions.append('peringat dan monitoring akses VLE')
    if 'assessment' in reasons: actions.append('pendampingan assessment')
    if row['risk_signal_count'] >= 3: actions.append('konseling atau tindak lanjut dosen wali')
    return '; '.join(actions) if actions else 'monitoring rutin'
result['recommended_action'] = result.apply(recommendation, axis=1)
return result

test_output = dataset.iloc[test_idx].copy().reset_index(drop=True)
test_output = add_knowledge_layer(test_output, best_pred, best_prob, thresholds)
print('Threshold kuartil bawah dari data train:')
display(pd.Series(thresholds, name='threshold').to_frame().round(2))
display(test_output['knowledge_risk_level'].value_counts().to_frame('jumlah'))
display(test_output[['id_student', 'code_module', 'code_presentation', 'probability_atrisk',
    'knowledge_risk_level', 'risk_reasons', 'recommended_action']].head())

```

Threshold kuartil bawah dari data train:

							threshold
assessment_score_mean							0.0
assessment_count							0.0
vle_total_clicks							47.0
vle_active_days							4.0
							jumlah
knowledge_risk_level							
Low Risk							2682
Medium Risk							1994
High Risk							1795
	id_student	code_module	code_presentation	probability_atrisk	knowledge_risk_level	risk_reasons	recommended_action
0	11391	AAA	2013J	0.340	Low Risk	Tidak ada sinyal aturan	monitoring rutin
1	38053	AAA	2013J	0.220	Low Risk	Tidak ada sinyal aturan	monitoring rutin
2	52130	AAA	2013J	0.120	Low Risk	Tidak ada sinyal aturan	monitoring rutin
3	63400	AAA	2013J	0.108	Low Risk	Tidak ada sinyal aturan	monitoring rutin
4	74372	AAA	2013J	0.576	Medium Risk	Total klik VLE rendah	peringat dan monitoring akses VLE

14. Evaluasi sistem gabungan

Tujuan: mengukur konsekuensi knowledge layer terhadap deteksi risiko.

Langkah: High Risk dan Medium Risk dipetakan menjadi AtRisk, lalu metriknya dibandingkan dengan model machine learning terbaik.

Output: perubahan recall dan precision setelah aturan ditambahkan.

```

knowledge_pred = test_output['knowledge_risk_level'].isin(['High Risk', 'Medium Risk']).astype(int)
comparison = pd.DataFrame([
    {'sistem': f'Model ML ({best_model_name})', **calculate_metrics(y_test, best_pred, best_prob)},
    {'sistem': 'Model + Knowledge Layer',
     'accuracy': accuracy_score(y_test, knowledge_pred),
     'precision_atrisk': precision_score(y_test, knowledge_pred, zero_division=0),
     'recall_atrisk': recall_score(y_test, knowledge_pred, zero_division=0),
     'f1_atrisk': f1_score(y_test, knowledge_pred, zero_division=0),
     'roc_auc': np.nan}
]).set_index('sistem')
display(comparison.style.format('{:.4f}', na_rep='-'))
print('Confusion matrix knowledge layer:')
display(pd.DataFrame(confusion_matrix(y_test, knowledge_pred),
    index=['Aktual Successful', 'Aktual AtRisk'],
    columns=['Prediksi Successful', 'Prediksi AtRisk']))

```

sistem	accuracy	precision_atrisk	recall_atrisk	f1_atrisk	roc_auc
Model ML (Random Forest)	0.7592	0.8032	0.7172	0.7578	0.8396
Model + Knowledge Layer	0.7136	0.7039	0.7849	0.7422	-

Confusion matrix knowledge layer:

	Prediksi Successful	Prediksi AtRisk
Aktual Successful	1951	1122
Aktual AtRisk	731	2667

15. Dashboard statis DVBI

Tujuan: menyatukan indikator model dan risiko menjadi tampilan monitoring akademik.

Langkah: menampilkan KPI, distribusi level, risiko per module-presentation, probabilitas, sinyal dominan, perilaku berdasarkan level, dan confusion matrix.

Output: area dengan tingkat risiko tertinggi dan faktor yang membutuhkan intervensi.

```

level_order = ['High Risk', 'Medium Risk', 'Low Risk']
level_colors = {'High Risk': '#D62828', 'Medium Risk': '#F4A261', 'Low Risk': '#2A9D8F'}
level_counts = test_output['knowledge_risk_level'].value_counts().reindex(level_order, fill_value=0)
module_risk = (test_output.assign(is_priority=test_output['knowledge_risk_level'].isin(['High Risk', 'Medium Risk'])))
    .groupby(['code_module', 'code_presentation'])['is_priority'].mean()
    .sort_values(ascending=False).head(10).mul(100)
all_signals = (test_output['risk_reasons'].str.get_dummies(sep=';')
    .drop(columns=['Tidak ada sinyal aturan'], errors='ignore').sum().sort_values(ascending=False))
behavior_level = test_output.groupby('knowledge_risk_level')[
    ['vle_total_clicks', 'assessment_score_mean']].median().reindex(level_order)

total_students = test_output['id_student'].nunique()
priority_count = test_output['knowledge_risk_level'].isin(['High Risk', 'Medium Risk']).sum()
priority_pct = 100 * priority_count / len(test_output)

fig = plt.figure(figsize=(20, 18), constrained_layout=True)
grid = fig.add_gridspec(4, 3, height_ratios=[0.45, 1.5, 1.5, 1.5])
fig.suptitle('DASHBOARD EARLY WARNING OULAD - MINGGU KE-4', fontsize=22, fontweight='bold')

kpis = [(f'{total_students:,}', 'Mahasiswa unik'), (f'{priority_count:,}', 'Prioritas intervensi'),
    (f'{priority_pct:.1f}%', 'Persentase prioritas')]
for i, (value, label) in enumerate(kpis):
    ax = fig.add_subplot(grid[0, i]); ax.axis('off')
    ax.text(0.5, 0.62, value, ha='center', va='center', fontsize=27, fontweight='bold', color='#264653')
    ax.text(0.5, 0.18, label, ha='center', va='center', fontsize=12)
    ax.add_patch(plt.Rectangle((0.02, 0.02), 0.96, 0.96, fill=False, edgecolor='#A8DADC', linewidth=2))

ax1 = fig.add_subplot(grid[1, 0])
ax1.pie(level_counts, labels=level_counts.index, autopct='%1.1f%%', startangle=90,
    colors=[level_colors[x] for x in level_counts.index])
ax1.set_title('Distribusi Level Risiko')

ax2 = fig.add_subplot(grid[1, 1])
module_risk.sort_values().plot(kind='barh', ax=ax2, color='#E76F51')
ax2.set_title('10 Module-Presentation dengan Prioritas Tertinggi')
ax2.set_xlabel('High + Medium Risk (%)'); ax2.set_ylabel('Module-Presentation')

ax3 = fig.add_subplot(grid[2, 0])
sns.histplot(data=test_output, x='probability_atrisk', hue='knowledge_risk_level',
    hue_order=level_order, palette=level_colors, bins=20, ax=ax3, element='step')
ax3.set_title('Distribusi Probabilitas AtRisk')

ax4 = fig.add_subplot(grid[2, 1])
all_signals.sort_values().plot(kind='barh', ax=ax4, color='#F4A261')
ax4.set_title('Sinyal Risiko Dominan'); ax4.set_xlabel('Jumlah mahasiswa')

ax5 = fig.add_subplot(grid[2, 2])
behavior_level.plot(kind='bar', ax=ax5, secondary_y='assessment_score_mean', color=['#457B9D', '#E9C46A'])
ax5.set_title('Median Perilaku per Level Risiko'); ax5.tick_params(axis='x', rotation=15)

ax6 = fig.add_subplot(grid[3, 0])
sns.heatmap(cm_best, annot=True, fmt='d', cmap='Blues', ax=ax6,
    xticklabels=['Successful', 'AtRisk'], yticklabels=['Successful', 'AtRisk'])
ax6.set_title(f'Confusion Matrix {best_model_name}'); ax6.set_xlabel('Prediksi'); ax6.set_ylabel('Aktual')

ax7 = fig.add_subplot(grid[3, 1])
metric_plot = comparison[['precision_atrisk', 'recall_atrisk', 'f1_atrisk']]
metric_plot.plot(kind='bar', ax=ax7, color=['#457B9D', '#E76F51', '#2A9D8F'])
ax7.set_title('Model ML vs Knowledge Layer'); ax7.set_ylim(0, 1); ax7.tick_params(axis='x', rotation=0)

```

```
ax7.legend(loc='lower right')
plt.show()
```

DASHBOARD EARLY WARNING OULAD — MINGGU KE-4



16. Daftar prioritas dan insight Business Intelligence

Tujuan: menghasilkan informasi yang langsung dapat ditindaklanjuti.

Langkah: mengurutkan mahasiswa High Risk berdasarkan probabilitas dan jumlah sinyal, lalu mencetak insight module-presentation serta sinyal dominan secara otomatis.

Output: 20 kasus teratas, alasan risiko, dan rekomendasi tindakan.

```
priority_table = (test_output[test_output['knowledge_risk_level'].isin(['High Risk', 'Medium Risk'])]
                  .sort_values(['knowledge_risk_level', 'probability_atrisk', 'risk_signal_count'],
                               ascending=[True, False, False])
                  [['id_student', 'code_module', 'code_presentation', 'probability_atrisk',
                    'knowledge_risk_level', 'risk_signal_count', 'risk_reasons', 'recommended_action']])

top_context = module_risk.index[0]
top_signal = all_signals.index[0]
print('INSIGHT BUSINESS INTELLIGENCE')
print(f'1. Prioritas tertinggi: module {top_context[0]} presentation {top_context[1]} '
      f'dengan {module_risk.iloc[0].1f}% kasus High/Medium Risk.')
print(f'2. Sinyal risiko paling dominan: {top_signal} ({int(all_signals.iloc[0].1f)} kasus).')
print(f'3. Total antrean intervensi: {len(priority_table)} student-module-presentation.')
print('4. Gunakan alasan risiko bersama probabilitas untuk memilih intervensi.')
print('5. Hasil ini adalah decision support; validasi manusia tetap wajib sebelum tindakan.')

display(priority_table.head(20).style.format({'probability_atrisk': '{:.1%}'))
```

INSIGHT BUSINESS INTELLIGENCE

1. Prioritas tertinggi: module GGG presentation 2014J dengan 100.0% kasus High/Medium Risk.
2. Sinyal risiko paling dominan: Skor assessment rendah (2,341 kasus).
3. Total antrean intervensi: 3,789 student-module-presentation.
4. Gunakan alasan risiko bersama probabilitas untuk memilih intervensi.
5. Hasil ini adalah decision support; validasi manusia tetap wajib sebelum tindakan.

	id_student	code_module	code_presentat ion	probability_atr isk	knowledge_risk _level	risk_signal_co unt	risk_reasons	recommended_ action
170	87604	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
171	88580	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
180	185240	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
186	230348	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
187	258402	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
189	269289	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
201	323914	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
209	353918	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
211	357038	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
220	372624	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
221	375312	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
290	505123	BBB	2013B	100.0%	High Risk	4	Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali

	id_student	code_module	code_presentat ion	probability_atr isk	knowledge_risk _level	risk_signal_co unt	risk_reasons	recommended_ action
	302	512924	BBB	2013B	100.0%	High Risk	4 Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
	331	532142	BBB	2013B	100.0%	High Risk	4 Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
	332	532317	BBB	2013B	100.0%	High Risk	4 Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
	345	535303	BBB	2013B	100.0%	High Risk	4 Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
	357	539426	BBB	2013B	100.0%	High Risk	4 Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
	366	541573	BBB	2013B	100.0%	High Risk	4 Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
	374	542760	BBB	2013B	100.0%	High Risk	4 Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali
	376	543864	BBB	2013B	100.0%	High Risk	4 Skor assessment rendah; Partisipasi assessment rendah; Total klik VLE rendah; Hari aktif VLE rendah	peringat dan monitoring akses VLE; pendampingan assessment; konseling atau tindak lanjut dosen wali

17. Ekspor hasil untuk pelaporan

Tujuan: menyimpan keluaran yang dapat digunakan dalam laporan atau analisis lanjutan.

Langkah: menyimpan metrik model dan daftar prioritas sebagai CSV di runtime Colab.

Output: lokasi file hasil.

```
OUTPUT_DIR = Path('/content/oulad_dvbi_output')
OUTPUT_DIR.mkdir(exist_ok=True)
test_results.reset_index().to_csv(OUTPUT_DIR / 'model_evaluation.csv', index=False)
comparison.reset_index().to_csv(OUTPUT_DIR / 'knowledge_layer_evaluation.csv', index=False)
priority_table.to_csv(OUTPUT_DIR / 'intervention_priority.csv', index=False)
print('Output tersimpan di:', OUTPUT_DIR)
print([p.name for p in OUTPUT_DIR.iterdir()])
```

```
Output tersimpan di: /content/oulad_dvbi_output
['intervention_priority.csv', 'model_evaluation.csv', 'knowledge_layer_evaluation.csv']
```

18. Kesimpulan dan keterbatasan

Kesimpulan: notebook membandingkan tiga model pada skenario early warning minggu ke-4. Model dipilih berdasarkan recall `AtRisk`, lalu knowledge-based layer menambahkan level risiko, alasan, dan rekomendasi. Dashboard menerjemahkan keluaran tersebut menjadi monitoring dan prioritas intervensi.

Keterbatasan:

- Label berasal dari hasil akhir, sedangkan prediktor dibatasi sampai hari ke-28.
- Threshold kuartil merupakan baseline berbasis data dan perlu divalidasi pakar akademik.
- Recall tinggi dapat meningkatkan false positive dan beban intervensi.
- OULAD berasal dari konteks Open University di Inggris sehingga penerapan pada institusi lain memerlukan validasi ulang.
- Feature importance menunjukkan kontribusi prediktif dan ditafsirkan sebagai asosiasi dalam model.